ESILV - Tutorial #4

Commodity Markets and Models

- This session (worth **5%**) lasts only 1h30.
- Solve exercises **on paper** and hand it in to your tutorial leader (*one per group*) before the end. (no email) – (no late submission)
- Ask your instructor for advice. Corrections will not be posted on DVL.
- In case of absence to this tutorial, you will be graded as zero. **Your instructor does not need to know the reason of your absence.** Official absence needs to be recorded directly with the scholar office (`scolarite-esilv@devinci.fr`).

Problem 1: `Gibson-Schwartz & Schwartz-Smith models for Copper Price Analysis`

Background

You are analyzing the copper commodity market using two prominent stochastic models: the Gibson-Schwartz (1990) two-factor model and the Schwartz-Smith (2000) short-term/long-term model. The reference articles are:

a) Schwartz, E., & Smith, J. E. (2000). Short-term variations and long-term dynamics in commodity prices. *Management Science*, 46(7), 893-911.

b) Gibson, R., & Schwartz, E. S. (1990). Stochastic convenience yield and the pricing of oil contingent claims. *The Journal of Finance*, 45(3), 959-976.

Figure 1 shows the Kalman filter estimation results for copper prices from 2010-2026, while Table 1 presents the estimated parameters.

Questions

Question 1: Model Specification (20%)

The Schwartz-Smith (2000) model decomposes the log spot price $\ln(S_t)$ into two components:

$$\ln(S_t) = \chi_t + \xi_t$$

where χ_t follows an Ornstein-Uhlenbeck process:

$$d\chi_t = -\kappa\chi_t dt + \sigma_\chi dZ_\chi$$

and ξ_t follows a Brownian motion with drift:

$$d\xi_t = \mu_\xi dt + \sigma_\xi dZ_\xi$$

with $dZ_\chi dZ_\xi = \rho dt$.

a) **(5%)** Interpret the economic meaning of χ_t and ξ_t in the context of copper prices. What real-world phenomena might each component represent?

b) **(5%)** The "half-life" of short-term deviations is defined as $t_{1/2} = -\ln(0.5)/\kappa$. Calculate the half-life using both the estimated value ($\kappa = 0.0148$) and true value ($\kappa = 1.2500$) from Table 1. Comment on the economic interpretation of these different half-lives.

c) **(5%)** In Figure 1, why does the equilibrium price (dark red line) appear smoother than the spot price (teal line)? What does this imply about the relative volatilities σ_χ and σ_ξ?

d) **(5%)** The lower panel of Figure 1 shows short-term deviations (χ_t) with green areas for positive deviations and red for negative. In what periods does copper appear to be overvalued relative to equilibrium? What might cause such persistent positive deviations?

Question 2: Parameter Interpretation and Estimation (25%)

Refer to Table 1 for parameter values.

a) **(10%)** The Kalman filter was used to estimate the unobservable state variables (χ_t and ξ_t) from futures prices. Explain the intuition behind using futures prices of different maturities to extract these two factors. Why would long-maturity futures primarily reflect ξ_t, while the difference between short and long-maturity futures reflects χ_t?

b) **(7%)** Compare the estimated values of σ_χ and σ_ξ with their true values. The estimation error for κ is particularly large (0.0148 vs 1.2500). Referring to Schwartz-Smith Section 6.1, explain why κ and the risk premium parameters are difficult to estimate precisely from futures data alone.

c) **(8%)** Using the conversion formulas from Schwartz-Smith Table 1, verify the relationship between the models. The Gibson-Schwartz model parameters can be expressed as:

$$\sigma_1 = \sqrt{\sigma_\xi^2 + \frac{\sigma_\chi^2}{\kappa^2} - \frac{2\rho\sigma_\chi\sigma_\xi}{\kappa}} \quad \text{and} \quad \sigma_2 = \kappa\sigma_\chi$$

Calculate σ_2 using both estimated and true parameters. Why is the estimated σ_2 so small (0.0038) compared to what you might expect for copper convenience yield volatility?

Question 3: Economic Interpretation and Trading Implications (30%)

a) **(10%)** The risk-neutral process in Schwartz-Smith includes adjustment terms λ_χ and λ_ξ:

$$d\chi_t = (-\kappa\chi_t - \lambda_\chi)dt + \sigma_\chi dZ_\chi^* \quad \text{and} \quad d\xi_t = (\mu_\xi - \lambda_\xi)dt + \sigma_\xi dZ_\xi^*$$

Explain the economic meaning of these risk premiums. How would positive values of λ_χ and λ_ξ affect the relationship between spot, equilibrium, and futures prices?

b) **(10%)** Consider a copper mining company evaluating two projects:

- Project A: Short-term development (lag: 0 years, decline: 40% per year)
- Project B: Long-term mine development (lag: 3 years, decline: 5% per year)

Based on Schwartz-Smith Section 7, which project's value is more sensitive to short-term deviations (χ_t)? Justify your answer using the concept of "half-life."

c) **(10%)** The volatility of a futures contract with maturity T in the Schwartz-Smith model is:

$$\sigma_F(T) = \sqrt{e^{-2\kappa T}\sigma_\chi^2 + \sigma_\xi^2 + 2e^{-\kappa T}\rho\sigma_\chi\sigma_\xi}$$

(a) Calculate $\sigma_F(0)$ and $\sigma_F(\infty)$ using the estimated parameters (assume $\rho = 0.3$ for calculations).

(b) Sketch the term structure of volatility. What does this imply for options on near-term vs. long-term copper futures?

(c) How does this relate to Samuelson's (1965) hypothesis?

Question 4: Model Comparison and Limitations (25%)

a) **(10%)** Based on the articles, explain why the short-term/long-term decomposition is more intuitive for traders than convenience yields. How does the "orthogonality" of factors in Schwartz-Smith simplify long-term investment analysis?

b) **(7%)** Discuss the Kalman filter's precision. How does measurement error in futures prices (bid-ask spreads, etc.) affect the state estimates, and why might the filtered estimates in Figure 1 appear smoother than the true states?

c) **(8%)** Suggest two extensions to the Schwartz-Smith model to better capture copper dynamics (2010-2026), such as regime-switching or jumps. Explain what additional parameters each would introduce.

End of Exercise. Please hand in your exercise on paper to your instructor.

Parameter	Description	Estimated Value	True Value
κ	Mean Reversion Speed	0.0148	1.2500
σ_χ	Short-term Volatility	0.2594	0.2800
σ_ξ	Equilibrium Volatility	0.1206	0.1600
ρ	Correlation	0.3000	0.3000
σ_1 (GS)	Spot Volatility	0.2448	N/A
σ_2 (GS)	Convenience Yield Vol	0.0038	N/A

Table 1: Estimated Parameters for Schwartz-Smith and Gibson-Schwartz Models (Copper 2010-2026)

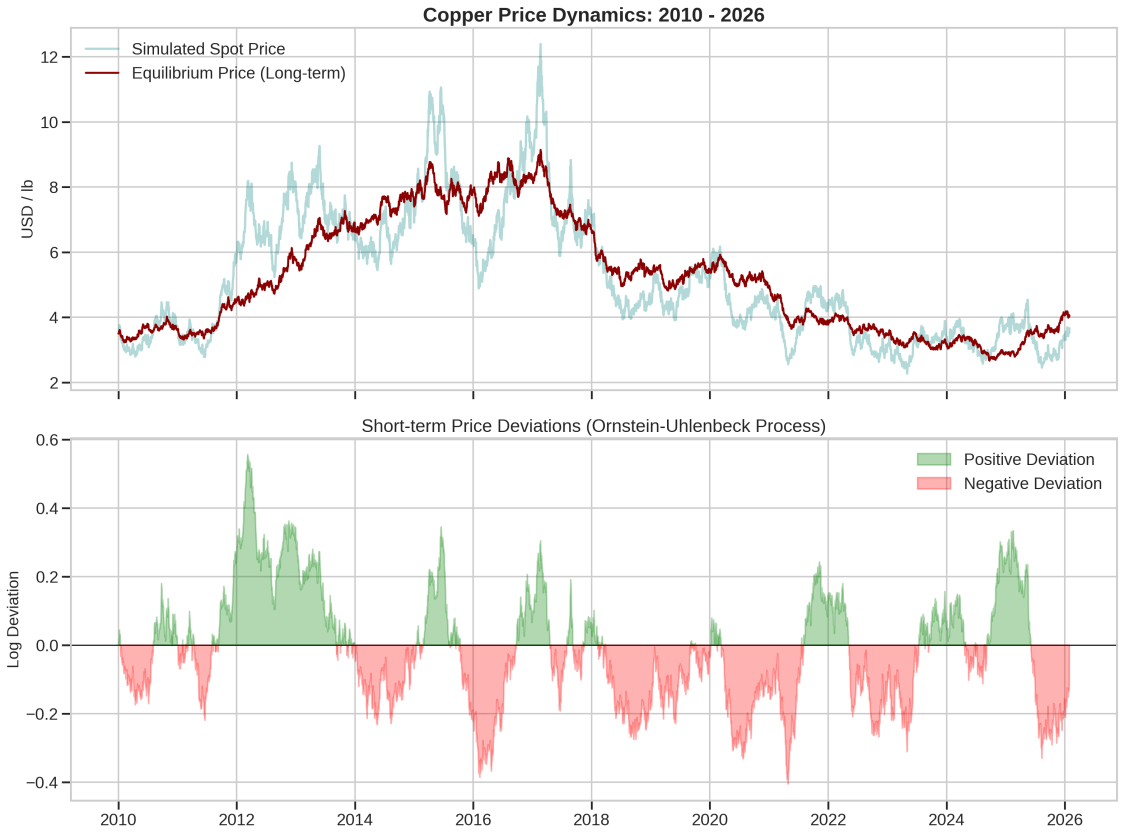

Figure 1: Kalman filter visualization of Gibson-Schwartz / Schwartz-Smith models for copper